

MODEL	ARCH.	LAYER	ACC.
S4	No gate	S4	18.3
-	No gate	S6	97.0
H3	H3	S4	57.0
Hyena	H3	Hyena	30.1
-	H3	S6	99.7
-	Mamba	S4	56.4
-	Mamba	Hyena	28.4
Mamba	Mamba	S6	99.8

Table 1: **(Selective Copying.)** Accuracy for combinations of architectures and inner sequence layers.

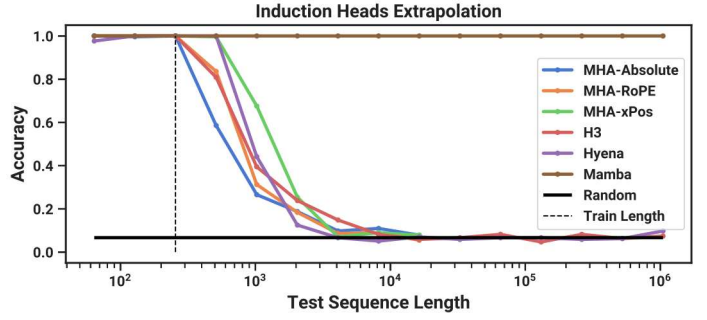


Table 2: **(Induction Heads.)** Models are trained on sequence length $2^8 = 256$, and tested on increasing sequence lengths of $2^6 = 64$ up to $2^{20} = 1048576$. Full numbers in Table 11.

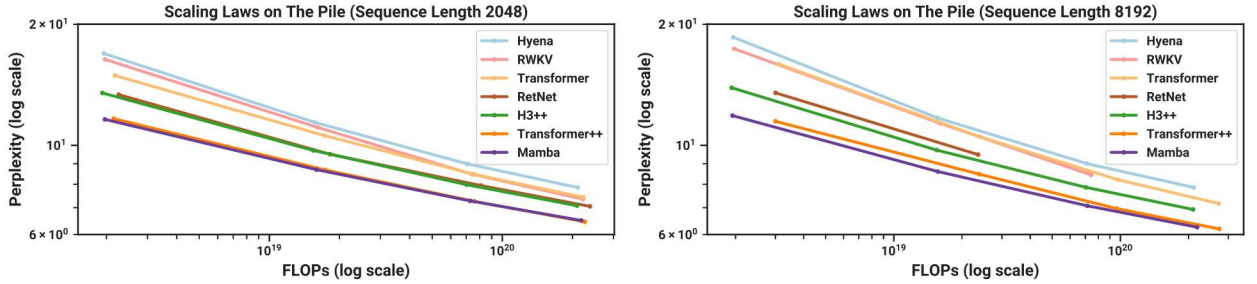


Figure 4: **(Scaling Laws.)** Models of size $\approx 125M$ to $\approx 1.3B$ parameters, trained on the Pile. Mamba scales better than all other attention-free models and is the first to match the performance of a very strong “Transformer++” recipe that has now become standard, particularly as the sequence length grows.

Out of positional encoding variants for attention models, xPos (which was designed for length extrapolation) is slightly better than the others; also note that all attention models were only tested up to sequence length $2^{14} = 16384$ due to memory limitations. Out of other SSMs, H3 and Hyena are similar, contrary to the findings in Poli et al. (2023).

4.2 Language Modeling

We evaluate the Mamba architecture on standard autoregressive language modeling against other architectures, on both pretraining metrics (perplexity) and zero-shot evaluations. We set the model sizes (depth and width) to mirror GPT3 specifications. We use the Pile dataset (L. Gao, Biderman, et al. 2020), and follow the training recipe described in Brown et al. (2020). All training details are in Appendix E.2.

4.2.1 Scaling Laws

For baselines, we compare against the standard Transformer architecture (GPT3 architecture), as well as the strongest Transformer recipe we know of (here referred to as Transformer++), based on the PaLM and LLaMa architectures (e.g. rotary embedding, SwiGLU MLP, RMSNorm instead of LayerNorm, no linear bias, and higher learning rates). We also compare against other recent subquadratic architectures (Figure 4). All model details are in Appendix E.2.

Figure 4 shows scaling laws under the standard Chinchilla (Hoffmann et al. 2022) protocol, on models from $\approx 125M$ to $\approx 1.3B$ parameters. **Mamba is the first attention-free model to match the performance of a very strong Transformer recipe (Transformer++) that has now become standard, particularly as the sequence length grows.** (We note that full results on context length 8k are missing for the RWKV and RetNet baselines, prior strong recurrent models that can also be interpreted as SSMs, because of a lack of efficient implementations leading to out-of-memory or unrealistic computation requirements.)